

Quiz 3

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given a system of equations

$$\begin{aligned}x + 2y + 3z &= 1 \\2x + 4y - z &= 4 \\3y + z &= 3\end{aligned}$$

(a) Write down the augmented matrix $[A, b]$ and which one is the first pivot?

$$[A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{array} \right]$$

first pivot: 1

(b) Find the *Elimination matrix* E_{21} which can be used to eliminate the first component below the first pivot and calculate $E_{21}[A, b]$.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}E_{21}[A, b] &= \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 0 & -7 & 2 \\ 0 & 3 & 1 & 3 \end{bmatrix}\end{aligned}$$

(c) Find the *Permutation matrix* P_{23} such that $P_{23}E_{21}[A, b]$ becomes upper triangular.

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P_{23}E_{21}[A, b] = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 3 & 1 & 3 \\ 0 & 0 & -7 & 2 \end{bmatrix}$$

Problem 2. (5 points) Let

$$A = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 1 & -2 \end{bmatrix}$$

(a) Compute AB and A^2 . Why is BA not valid?

$$AB = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 3 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 12 \\ 6 & 4 & -5 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -18 \\ 6 & 13 \end{bmatrix}$$

BA is not valid since ^{the} number of columns of B is 3 and the number of rows of A is 2, they don't match.

(b) Find A^{-1} .

$$\begin{aligned} A^{-1} &= \frac{1}{2 \cdot 4 - (-3) \cdot 1} \begin{bmatrix} 4 & 3 \\ -1 & 2 \end{bmatrix} \\ &= \frac{1}{11} \begin{bmatrix} 4 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{11} & \frac{3}{11} \\ -\frac{1}{11} & \frac{2}{11} \end{bmatrix} \end{aligned}$$